

1. Administrative & Study Identification

Official Title: A Phase III Multicenter, Randomized, Double-Blind, Double-Dummy, Parallel-Group Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Ocrelizumab in Adult Patients With Primary Progressive Multiple Sclerosis **Study Title:** A Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Ocrelizumab in Adult Participants With Primary Progressive Multiple Sclerosis **Short Title/Acronym:** FENTrepid **Protocol Number:** GN41791 **NCT Number:** NCT04544449 **Posting ID:** 1071196450612017369 **Trial Status:** Active, Not Recruiting **Sponsor/Company Name:** Roche **Investigational Products:** Fenebrutinib, Ocrelizumab (Ocrevus) **Phase of Development:** Phase 3 **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of fenebrutinib on disability progression in adult participants with Primary Progressive Multiple Sclerosis (PPMS) compared to ocrelizumab. **Secondary Objectives:** To evaluate the effect of fenebrutinib on longer-term composite and standard disability progression metrics compared to the standard of care.

2.2 Background & Rationale Primary progressive multiple sclerosis (PPMS) is a form of MS characterized by the continuous worsening of disability from symptom onset, independent of relapse activity. While ocrelizumab is an approved therapy for PPMS, a significant unmet medical need remains. Fenebrutinib is a noncovalent, reversible, and central nervous system (CNS)-penetrant Bruton’s tyrosine kinase (BTK) inhibitor designed to cross the blood-brain barrier. By targeting both peripheral B cells and CNS microglia, it aims to address the chronic damage in the brain that drives long-term disability progression more effectively.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Ocrelizumab) **Blinding:** Double-Blind, Double-Dummy **Randomization:** 1:1 Parallel-Group via interactive voice or web-based response system (IxRS)

3.2 Planned Enrollment & Duration Targeted Enrollment: 985 adult participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 2020 **Study Duration:** At least 120 weeks of double-

blind treatment. The Open-Label Extension (OLE) phase is contingent on a positive benefit-risk result in the Primary Analysis of the study. Participants who discontinue study medication early or discontinue from the study will not be replaced.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults aged 18 to 65 years (restricted to 46-65 years for sites in Germany and Italy only). 2. A diagnosis of PPMS in accordance with the revised 2017 McDonald Criteria. 3. Expanded Disability Status Scale (EDSS) score from 3.0 to 6.5 inclusive at screening. 4. Documented disability progression in the 12 months prior to screening. 5. Pyramidal functional subscore ≥ 2 at screening. 6. For participants currently receiving proton pump inhibitors (PPIs), H2-receptor antagonists (H2RAs), symptomatic treatment for MS (e.g. fampridine, cannabis) and/or physiotherapy: treatment at a stable dose during the screening period prior to initiation and plans to remain at a stable dose. 7. Neurologically stable for at least 30 days prior to randomization and baseline assessments. 8. Ability to complete the 9-Hole Peg Test (9-HPT) for each hand in <240 seconds. 9. Ability to perform Timed 25-Foot Walk Test (T25FWT) in <150 seconds. 10. For female participants of childbearing potential: agreement to remain abstinent or use contraceptive measures, and refrain from donating eggs. 11. For male participants: agreement to remain abstinent or use contraceptive measures, and refrain from donating sperm.

OLE Inclusion Criteria: 1. Completed the Double-Blind Treatment (DBT) phase of the study (remaining on study treatment; no other DMT administered) and who, in the opinion of the investigator, may benefit from treatment with fenebrutinib. 2. Contraception and abstinence agreements for both male and female participants as described above.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding women, intending to become pregnant during the study, or within 6 to 12 months after the final dose of study drug. Male participants intending to father a child during the study or for 28 days after final dose. 2. For participants enrolled in Germany and Italy only: Presence of gadolinium-enhancing lesions on T1-weighted MRI (T1Gd+) on the screening MRI. 3. Any known or suspected active infection at screening (excluding onychomycosis), including positive tests for Hepatitis B and C, active/latent/inadequately treated tuberculosis (TB), or confirmed/suspected progressive multifocal leukoencephalopathy (PML). 4. Previous history of a serious Infusion-Related Reaction (IRR) (CTCAE Grade ≥ 4) and/or hypersensitivity reaction to ocrelizumab. 5. History of cancer (hematologic or solid) within 10 years of screening (exceptions: cured basal/squamous cell skin carcinoma, in situ cervical carcinoma treated >1 year prior). 6. Presence of

cirrhosis (Child-Pugh Class A, B, or C). Chronic liver disease (unless stable for >6 months). Acute liver disease. 7. Concomitant disease requiring chronic systemic corticosteroids, immunosuppressants, or medication impacting primary evaluations. 8. History of alcohol or drug abuse within 12 months prior to screening. 9. Lack of peripheral venous access. 10. Previous treatment with immunomodulatory or immunosuppressive medication without appropriate washout. 11. Receipt of a live or live-attenuated vaccine within 6 weeks prior to randomization. 12. Immunocompromised state, primary/secondary immunodeficiency, or history of transplantation/antirejection therapy. 13. Known bleeding diathesis, anemia, or history of hospitalization/transfusion for gastrointestinal (GI) bleed. 14. Any previous treatment with cladribine, mitoxantrone, daclizumab, alemtuzumab, or cyclophosphamide. 15. Other neurological disorders interfering with MS diagnosis, or clinically significant cardiovascular, psychiatric, pulmonary, renal, hepatic, endocrine, metabolic, or gastrointestinal disease.

OLE Exclusion Criteria: 1. Chronic liver disease unless considered stable for > 6 months. 2. Acute liver disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Daily oral fenebrutinib (given alongside intravenous [IV] placebo matching ocrelizumab) OR Intravenous ocrelizumab (given alongside oral placebo matching fenebrutinib). **Route of Administration:** Oral and IV (Double-Dummy design). **Duration of Treatment:** At least 120 weeks of double-blind treatment, with an Open-Label Extension (OLE) phase contingent on a positive benefit-risk result.

5.2 Concomitant & Prohibited Therapy Permitted: Standard symptomatic management therapies for MS (e.g., fampridine, cannabis) provided they are on a stable dose. **Prohibited:** Concurrent administration of other Disease-Modifying Therapies (DMTs) during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to 6 weeks before V1 Informed Consent, EDSS assessment, MRI scans, baseline liver function & safety labs Baseline Day 0 Randomization (1:1), First Dose of Oral/IV study treatments via IxRS Interim Over 120 weeks Blinded safety & efficacy assessments, 9-HPT, T25FWT, EDSS scoring, Liver enzyme monitoring End of Study \$\\ge\$ Week 120 Final clinical functional assessments, transition to Open-Label Extension (OLE) if eligible

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Time to Onset of Composite 12-week Confirmed Disability Progression (cCDP12), a combined measure incorporating Expanded Disability Status Scale (EDSS) progression, timed 25-foot walk (T25FWT), and nine-hole peg test (9-HPT) performance.

7.2 Secondary Endpoints 1. Time to Onset of Composite 24-week CDP (cCDP24). 2. Time to Onset of 12-week CDP (CDP12). 3. Time to Onset of 24-week CDP (CDP24).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the trial for general AEs, focusing on infections, gastrointestinal issues (nausea), hemorrhage, and transient/reversible elevations in liver enzymes associated with BTK inhibitors. **Serious Adverse Events (SAEs):** Standard reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** Independent safety monitoring, especially crucial for continuous liver safety evaluation.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses. **Sample Size Justification:** Globally enrolled $N=985$ to adequately power the comparison of 12-week cCDP12 hazard ratios (fenebrutinib vs. ocrelizumab) over the 120+ week observation period. **Handling of Missing Data:** Standard survival analysis methods (e.g., Cox proportional hazards) for time-to-event endpoints; participants who discontinue study medication early or discontinue from the study will not be replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular oversight via onsite and remote monitoring to ensure the integrity of the double-dummy blinding and proper functional assessment. **Audit Trail:** Data collection managed securely through an interactive voice/web-based response system (IxRS) to preserve treatment masking.

11. Study Identifiers & Governance

Primary ID: GN41791 **Registry Number:** NCT04544449 **Posting ID:** 1071196450612017369 **Sponsor/Lead Organization:** Roche **Study Title:**

A Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Ocrelizumab in Adult Participants With Primary Progressive Multiple Sclerosis **Official Regulatory Title:** A Phase III Multicenter, Randomized, Double-Blind, Double-Dummy, Parallel-Group Study to Evaluate the Efficacy and Safety of Fenebrutinib Compared With Ocrelizumab in Adult Patients With Primary Progressive Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Primary Progressive Multiple Sclerosis (PPMS)
Disease Area: Multiple Sclerosis, Primary Progressive **Diagnosis Threshold:** Revised 2017 McDonald Criteria; EDSS 3.0 to 6.5
Medical Methodology: CNS-penetrant Bruton's tyrosine kinase (BTK) inhibition vs. anti-CD20 monoclonal antibody depletion
Optimization Target: Delaying 12-week composite confirmed disability progression (cCDP12)

13. Study Procedures & Methodology

Management Pathway: Double-dummy active comparator pathway (oral therapy + IV placebo vs. IV therapy + oral placebo). **Education Protocol (HCP):** Specialized training for independent rating physicians on administering the EDSS, T25FWT, and 9-HPT. **Education Protocol (Patient):** Guidance on managing both daily oral treatments and episodic IV infusion visits. **Quality Audit Mechanism:** Independent central review of disability progression metrics to confirm cCDP12.

14. Observation & Follow-Up Schedule

Total Duration: Minimum 120 weeks (Double-Blind Treatment phase)
Onsite Visit Cadence: Set clinical visits at baseline and regularly scheduled intervals through Week 120 to measure functional decline. **Remote Monitoring Frequency:** Intermittent follow-up for safety and oral drug adherence tracking. **Checklist Review Interval:** Regular laboratory checks targeted at monitoring transient liver enzyme levels.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				